

Numerical Analysis of Partial Differential Equations

Lecture Notes
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About these Lecture Notes

The present lecture notes accompanies the lecture Numerical Analysis of Partial Differential Equations, which I teach in the Summer Semester 2026 at Freie Universität Berlin. The notes are compiled from various sources and are largely based on lecture materials by Universität Münster (M. Ohlberger), Universität Heidelberg (R. Rannacher), Universität Basel (H. Harbrecht), Universität Stuttgart (B. Harrach), and Freie Universität Berlin (V. John).

These lecture notes are intended as an aid for reviewing the course material, but they cannot and are not meant to fully reproduce the exact wording of the lecture. In class, we will motivate definitions, theorems, and numerical algorithms, discuss their properties, and illustrate the results with examples.

Please send suggestions for improvements of any kind to: c.schillings@fu-berlin.de

Literatur zur Vorlesung:

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Contents

1. Introduction	7
1.1. Examples	7
1.1.1. Potential equation.	7
1.1.2. Heat equation.	8
1.1.3. Poisson equation.	9
1.1.4. Wave equation.	9
1.2. Classification	10
A. Notation and Theoretical Foundations	12
A.1. Differential Operators and Hölder Spaces	12
A.2. L^p spaces	14
A.3. Gauss theorem and integration by parts	16

1. Introduction

Many phenomena in physics, engineering, and the natural sciences can be described by partial differential equations expressing conservation laws and material relations. This chapter introduces basic model equations and develops the analytical tools needed to understand their properties and solutions.

1.1 Examples

The following examples illustrate how fundamental partial differential equations arise from physical principles such as conservation laws and constitutive relations. They demonstrate typical modelling steps and highlight the characteristic behaviour of different equation types.

1.1.1 Potential equation.

Let $\Omega \subset \mathbb{R}^2$ be a domain, that is, an open and connected set, and let $\Gamma := \partial\Omega$ denote its boundary. Suppose that the graph of a function $g : \Gamma \rightarrow \mathbb{R}$ describes a wire loop spanning a soap film. This soap film can be described by a function $u : \Omega \rightarrow \mathbb{R}$ whose shape has minimal surface area:

$$\int_{\Omega} \sqrt{1 + u_x^2 + u_y^2} \, dx \, dy \rightarrow \min.$$

Since

$$\sqrt{1 + z} = 1 + \frac{z}{2} + \mathcal{O}(z^2),$$

the integrand can be approximated for small values of u_x and u_y by

$$F(u) := \frac{1}{2} \int_{\Omega} (u_x^2 + u_y^2) \, dx \, dy \rightarrow \min.$$

If $u \in C^2(\Omega) \cap C(\bar{\Omega})$ with $u|_{\Gamma} = g$ is a solution of this minimization problem, then for every $v \in C^1(\Omega) \cap C(\bar{\Omega})$ with $v|_{\Gamma} = 0$ we obtain

$$0 = \lim_{\varepsilon \rightarrow 0} \frac{F(u + \varepsilon v) - F(u)}{\varepsilon} = \int_{\Omega} (u_x v_x + u_y v_y) \, dx \, dy = \int_{\Omega} \langle \nabla u, \nabla v \rangle \, d\mathbf{x}. \quad (1.1)$$

Gauss' theorem yields the identity

$$\int_{\Omega} \Delta u \, v \, d\mathbf{x} + \int_{\Omega} \langle \nabla u, \nabla v \rangle \, d\mathbf{x} = \int_{\Gamma} \underbrace{v}_{=0} \frac{\partial u}{\partial \mathbf{n}} \, d\sigma = 0,$$

where $\Delta u = u_{xx} + u_{yy}$ denotes the Laplace operator. Substituting this into (1.1), we obtain the condition

$$0 = \int_{\Omega} \Delta u \, v \, d\mathbf{x}$$

for all $v \in C^1(\Omega) \cap C(\bar{\Omega})$ with $v|_{\Gamma} = 0$. Hence, the function u must satisfy the potential equation, or Laplace equation,

$$\Delta u(\mathbf{x}) = 0, \quad \mathbf{x} \in \Omega. \quad (1.2)$$

A simple representation formula for the Laplace equation is available in the case of the disk

$$\Omega = \{\mathbf{x} \in \mathbb{R}^2 : \|\mathbf{x}\|_2 < 1\}.$$

→ *Exercise.*

1.1.2 Heat equation.

Let $\Omega \subset \mathbb{R}^d$ be a bounded domain, and let

$$u : \mathbb{R}_{\geq 0} \times \bar{\Omega} \rightarrow \mathbb{R}$$

describe the temperature distribution. At time $t = 0$, let the initial distribution

$$u(0, \mathbf{x}) = u_0(\mathbf{x}) \in C(\bar{\Omega})$$

be given. In addition, let a heat source

$$f \in C(\mathbb{R}_{>0} \times \Omega)$$

be prescribed in Ω , and let boundary values

$$g \in C(\mathbb{R}_{>0} \times \Gamma), \quad \Gamma = \partial\Omega,$$

be given on the boundary.

By the conservation law, for every control volume $V \subset \Omega$ we have

$$\underbrace{\int_V \frac{\partial}{\partial t} u(t, \mathbf{x}) \, d\mathbf{x}}_{\text{heat content in } V} = - \underbrace{\int_{\partial V} \langle \mathbf{q}(t, \mathbf{x}), \mathbf{n}(\mathbf{x}) \rangle \, d\sigma}_{\text{heat flux from outside}} + \underbrace{\int_V f(t, \mathbf{x}) \, d\mathbf{x}}_{\text{heat source}}.$$

According to the constitutive law, the heat flux satisfies

$$\mathbf{q}(t, \mathbf{x}) = -c(\mathbf{x}) \nabla u(t, \mathbf{x}),$$

where $c \geq c_0 > 0$ is the material-dependent heat conductivity. Substituting this into Gauss' theorem yields

$$- \int_{\partial V} \langle \mathbf{q}(t, \mathbf{x}), \mathbf{n}(\mathbf{x}) \rangle \, d\sigma = - \int_V \operatorname{div} \mathbf{q}(t, \mathbf{x}) \, d\mathbf{x} = \int_V \operatorname{div} (c(\mathbf{x}) \nabla u(t, \mathbf{x})) \, d\mathbf{x}.$$

Thus, for the temperature distribution we obtain for all control volumes V

$$\int_V \left\{ \frac{\partial}{\partial t} u(t, \mathbf{x}) - \operatorname{div} (c(\mathbf{x}) \nabla u(t, \mathbf{x})) \right\} d\mathbf{x} = \int_V f(t, \mathbf{x}) \, d\mathbf{x},$$

which means that

$$\frac{\partial}{\partial t} u(t, \mathbf{x}) - \operatorname{div} (c(\mathbf{x}) \nabla u(t, \mathbf{x})) = f(t, \mathbf{x}), \quad (t, \mathbf{x}) \in \mathbb{R}_{>0} \times \Omega.$$

If c is constant, for instance $c = 1$, then the temperature distribution

$$u \in C^2(\mathbb{R}_{>0} \times \Omega) \cap C(\mathbb{R}_{\geq 0} \times \bar{\Omega})$$

satisfies

$$\frac{\partial}{\partial t} u(t, \mathbf{x}) - \Delta u(t, \mathbf{x}) = f(t, \mathbf{x}), \quad (t, \mathbf{x}) \in \mathbb{R}_{\geq 0} \times \Omega. \quad (1.3)$$

1.1.3 Poisson equation.

If the data f and g in the heat equation are independent of time, then as $t \rightarrow \infty$ an equilibrium state is reached. This means that $\partial u / \partial t = 0$, and (1.3) reduces to the Poisson equation

$$-\Delta u(\mathbf{x}) = f(\mathbf{x}), \quad \mathbf{x} \in \Omega. \quad (1.4)$$

This equation is also of central importance in electrostatics: if the charge density $f : \Omega \rightarrow \mathbb{R}$ is known in Ω , then the electric potential u satisfies the Poisson equation there.

1.1.4 Wave equation.

The motion in an ideal gas is governed by three laws. As usual, the velocity is denoted by $\mathbf{v}$, the density by ρ , and the pressure by p .

1. *Continuity equation:*

$$\frac{\partial \rho}{\partial t} = -\rho_0 \operatorname{div} \mathbf{v}.$$

By conservation of mass, the change of mass in a control volume V equals the flux through its boundary, that is,

$$\int_{\partial V} \rho \langle \mathbf{v}, \mathbf{n} \rangle d\sigma.$$

Using Gauss' theorem, this yields

$$\frac{\partial \rho}{\partial t} = -\operatorname{div}(\rho \mathbf{v}).$$

Approximating ρ by a constant, time-independent density ρ_0 gives the equation above.

2. *Newton's law:*

$$\rho_0 \frac{\partial \mathbf{v}}{\partial t} = -\nabla p.$$

The pressure gradient induces a force field that causes the acceleration of the particles.

3. *Equation of state:*

$$p = c^2 \rho.$$

In an ideal gas at constant temperature, the pressure is proportional to the density.

These three laws imply

$$\frac{\partial^2 p}{\partial t^2} = c^2 \frac{\partial^2 \rho}{\partial t^2} = -c^2 \frac{\partial}{\partial t} (\rho_0 \operatorname{div} \mathbf{v}) = -c^2 \operatorname{div} \left(\rho_0 \frac{\partial \mathbf{v}}{\partial t} \right) = c^2 \operatorname{div}(\nabla p) = c^2 \Delta p.$$

Other examples leading to the wave equation

$$\frac{\partial^2 p}{\partial t^2}(t, \mathbf{x}) = c^2 \Delta p(t, \mathbf{x}), \quad (t, \mathbf{x}) \in \mathbb{R}_{>0} \times \Omega, \quad (1.5)$$

arise in two space dimensions for a vibrating membrane and in one space dimension for a vibrating string.

1.2 Classification

Let $\Omega \subset \mathbb{R}^d$ be a domain, and let $\mathcal{L} : C^2(\Omega) \rightarrow C(\Omega)$ be a general linear differential operator of second order,

$$(\mathcal{L}u)(\mathbf{x}) = - \sum_{i,j=1}^d a_{ij}(\mathbf{x}) \frac{\partial^2}{\partial x_i \partial x_j} u(\mathbf{x}) + \sum_{i=1}^d b_i(\mathbf{x}) \frac{\partial}{\partial x_i} u(\mathbf{x}) + c(\mathbf{x})u(\mathbf{x}), \quad (1.6)$$

where

$$\mathbf{A} = [a_{ij}]_{i,j=1}^d \in [C(\Omega)]^{d \times d}, \quad \mathbf{b} = [b_i]_{i=1}^d \in [C(\Omega)]^d, \quad c \in C(\Omega).$$

The associated differential equation is

$$(\mathcal{L}u)(\mathbf{x}) = f(\mathbf{x}), \quad \mathbf{x} \in \Omega. \quad (1.7)$$

Since for $u \in C^2(\Omega)$ the second derivatives are symmetric, i.e.,

$$\frac{\partial^2 u}{\partial x_i \partial x_j} = \frac{\partial^2 u}{\partial x_j \partial x_i},$$

we may assume without loss of generality that $a_{ij} = a_{ji}$. Hence, the matrix $\mathbf{A}$ is symmetric and therefore has only real eigenvalues. The operator

$$- \sum_{i,j=1}^d a_{ij} \frac{\partial^2}{\partial x_i \partial x_j}$$

is called the principal part of $\mathcal{L}$.

Definition 1.1 The differential operator (1.6) is called

- *elliptic* at $\mathbf{x}$ if all eigenvalues of $\mathbf{A}(\mathbf{x})$ are positive,
- *parabolic* at $\mathbf{x}$ if $d - 1$ eigenvalues of $\mathbf{A}(\mathbf{x})$ are positive and one eigenvalue is zero, but

$$\text{rank}([\mathbf{A}(\mathbf{x}), \mathbf{b}(\mathbf{x})]) = d,$$

- *hyperbolic* at $\mathbf{x}$ if $d - 1$ eigenvalues of $\mathbf{A}(\mathbf{x})$ are positive and one eigenvalue is negative.

The operator (1.6) is called elliptic, parabolic, or hyperbolic in Ω if it has the corresponding property for all $\mathbf{x} \in \Omega$. Accordingly, the differential equation (1.7) is called elliptic, parabolic, or hyperbolic if its associated differential operator has this property.

Example 1.2 The potential equation (1.2) and the Poisson equation (1.4) are elliptic, the heat equation (1.3) is parabolic, and the wave equation (1.5) is hyperbolic. $\rightarrow$ Exercise. $\triangle$

In addition to the differential equation (??), suitable initial or boundary conditions must be imposed in order to obtain a meaningful problem formulation.

Definition 1.3 A problem is called *well posed* if a solution exists, is unique, and depends continuously on the prescribed data. Otherwise, the problem is called *ill posed*.

The distinction between different types of partial differential equations is valuable because each type exhibits its own fundamentally unique and meaningful properties.

1. *Elliptic differential equations:* For elliptic problems, boundary conditions are prescribed: for given $f \in C(\Omega)$ and $g \in C(\Gamma)$, find $u \in C^2(\Omega) \cap C(\bar{\Omega})$ such that

$$\mathcal{L}u = f \quad \text{in } \Omega, \quad u = g \quad \text{on } \Gamma. \quad (1.8)$$

These boundary conditions are called *Dirichlet boundary conditions*. In practice, *Neumann boundary conditions*,

$$\frac{\partial u}{\partial \mathbf{n}} = g \quad \text{on } \Gamma,$$

also occur frequently. Solutions of elliptic differential equations satisfy the maximum principle.

2. *Parabolic differential equations:* Parabolic differential equations describe diffusion processes. The distinguished coordinate direction is usually time, so that the equation can often be written in the form

$$u_t + \mathcal{L}u = f,$$

where $\mathcal{L}$ is an elliptic differential operator. In addition, initial and boundary values are prescribed: for given

$$f \in C(\mathbb{R}_{>0} \times \Omega), \quad g \in C(\mathbb{R}_{>0} \times \Gamma), \quad u_0 \in C(\bar{\Omega}),$$

find

$$u \in C^2(\mathbb{R}_{>0} \times \Omega) \cap C(\mathbb{R}_{\geq 0} \times \bar{\Omega})$$

such that

$$\begin{aligned} u_t + \mathcal{L}u &= f \quad \text{in } \mathbb{R}_{>0} \times \Omega, \\ u &= g \quad \text{on } \mathbb{R}_{>0} \times \Gamma \quad (\text{boundary condition}), \\ u(0, \cdot) &= u_0 \quad \text{on } \bar{\Omega} \quad (\text{initial condition}). \end{aligned}$$

3. *Hyperbolic differential equations:* Here again, one coordinate is distinguished and can be interpreted as time. Hence the equation can often be written as

$$u_{tt} + \mathcal{L}u = f,$$

with an elliptic differential operator $\mathcal{L}$. Hyperbolic equations describe oscillatory phenomena. Meaningful problems are obtained by prescribing initial conditions: for given

$$f \in C(\mathbb{R}_{>0} \times \Omega), \quad g \in C(\mathbb{R}_{>0} \times \Gamma), \quad u_0, u_1 \in C(\bar{\Omega}),$$

find

$$u \in C^2(\mathbb{R}_{>0} \times \Omega) \cap C(\mathbb{R}_{\geq 0} \times \bar{\Omega})$$

such that

$$\begin{aligned} u_{tt} + \mathcal{L}u &= f \quad \text{in } \mathbb{R}_{>0} \times \Omega, \\ u &= g \quad \text{on } \mathbb{R}_{>0} \times \Gamma \quad (\text{boundary condition}), \\ u(0, \cdot) &= u_0, \quad u_t(0, \cdot) = u_1 \quad \text{on } \bar{\Omega} \quad (\text{initial conditions}). \end{aligned}$$

If the differential operator is invariant under rigid motions, that is, under translations and rotations, then the elliptic part $\mathcal{L}$ has the form

$$\mathcal{L}u = -a\Delta u + cu.$$

→ Exercise.

A. Notation and Theoretical Foundations

In this section we introduce notation and recall some fundamental results from analysis that we will repeatedly refer to throughout the lecture. All theorems are stated without proof. For a more detailed presentation we refer to the literature on analysis in several spatial dimensions.

A.1 Differential Operators and Hölder Spaces

We first introduce notation for differential operators and define the Hölder spaces $C^{m,\alpha}(\Omega)$ as a generalization of the spaces $C^m(\Omega)$ of m -times continuously differentiable functions.

Definition A.1 Let $u : \mathbb{R}^d \rightarrow \mathbb{R}$ be a scalar-valued function. Then $D^\alpha u$ denotes the partial derivative of order k of u if $|\alpha| = k$ and

$$\partial^\alpha u := \frac{\partial^\alpha u}{\partial x^\alpha} := \left(\frac{\partial}{\partial x_1} \right)^{\alpha_1} \cdots \left(\frac{\partial}{\partial x_d} \right)^{\alpha_d} u,$$

where $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$ is a multi-index with notation

$$|\alpha| := \sum_{i=1}^d \alpha_i, \quad x^\alpha := x_1^{\alpha_1} \cdots x_d^{\alpha_d}.$$

For functions $u(t, x)$, $(t, x) \in \mathbb{R} \times \mathbb{R}$, we use the abbreviations

$$\partial_t u(t, x) := \frac{\partial}{\partial t} u(t, x), \quad \partial_x u(t, x) := \frac{\partial}{\partial x} u(t, x).$$

For functions $u(x)$, $x \in \mathbb{R}^d$, we write

$$\partial_i u(x) := \partial_{x_i} u(x) := \frac{\partial}{\partial x_i} u(x).$$

Similarly, we abbreviate higher derivatives by $\partial_i^k u$.

Remark A.2 (Coordinate derivatives in low dimension) For $x = (x_1, x_2) \in \mathbb{R}^2$ we use the classical abbreviations

$$u_x := \partial_1 u, \quad u_y := \partial_2 u.$$

More generally, for $x \in \mathbb{R}^d$,

$$\partial_i u = \frac{\partial u}{\partial x_i}, \quad \nabla u = (\partial_1 u, \dots, \partial_d u)^\top.$$

△

Definition A.3 Gradient, divergence, curl and Laplace operator

Let $u : \mathbb{R}^d \rightarrow \mathbb{R}$ and $\mathbf{v} : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be sufficiently smooth functions.

The gradient is defined by

$$\nabla u(x) := \text{grad } u(x) := (\partial_1 u(x), \dots, \partial_d u(x))^\top.$$

The divergence is defined by

$$\nabla \cdot \mathbf{v}(x) := \text{div } \mathbf{v}(x) := \sum_{i=1}^d \partial_i v_i(x).$$

The Laplace operator is defined by

$$\Delta u(x) := \nabla \cdot (\nabla u(x)) = \sum_{i=1}^d \partial_i^2 u(x).$$

For $d = 3$, the curl is defined as

$$\nabla \times \mathbf{v} = \text{rot } \mathbf{v} = \begin{pmatrix} \partial_2 v_3 - \partial_3 v_2 \\ \partial_3 v_1 - \partial_1 v_3 \\ \partial_1 v_2 - \partial_2 v_1 \end{pmatrix}.$$

Definition A.4 (Hölder spaces) Let $\Omega \subset \mathbb{R}^d$ be open and bounded.

We denote by

$$C(\Omega, \mathbb{R}^n) := C^0(\Omega, \mathbb{R}^n)$$

the space of continuous functions on Ω .

We define

$$C(\overline{\Omega}, \mathbb{R}^n) := \{u : \Omega \rightarrow \mathbb{R}^n \mid u \text{ admits a continuous extension to } \overline{\Omega}\}.$$

For $m \in \mathbb{N}$ we define

$$C^m(\Omega, \mathbb{R}^n) := \{u \in C(\Omega, \mathbb{R}^n) \mid \partial^\beta u \in C(\Omega, \mathbb{R}^n) \text{ for all } |\beta| \leq m\}.$$

Similarly,

$$C^m(\overline{\Omega}, \mathbb{R}^n) := \{u \in C^m(\Omega, \mathbb{R}^n) \mid \partial^\beta u \text{ admits a continuous extension to } \overline{\Omega} \text{ for all } |\beta| \leq m\}.$$

For $m = 0$, we often write $C(\Omega, \mathbb{R}^n)$ instead of $C^0(\Omega, \mathbb{R}^n)$ (and $C(\overline{\Omega}, \mathbb{R}^n)$ accordingly).

For $n = 1$ we simply write

$$C(\Omega), \quad C(\overline{\Omega}), \quad C^m(\Omega), \quad C^m(\overline{\Omega}).$$

For $u \in C(\overline{\Omega})$ and $\alpha \in [0, 1]$ we define the Hölder seminorm

$$\text{hol}_\alpha(u, \overline{\Omega}) := \sup_{\substack{x, y \in \overline{\Omega} \\ x \neq y}} \frac{|u(x) - u(y)|}{|x - y|^\alpha}.$$

The Hölder spaces are defined by

$$C^{m, \alpha}(\overline{\Omega}) := \left\{ u \in C^m(\overline{\Omega}) \mid \text{hol}_\alpha(\partial^\beta u, \overline{\Omega}) < \infty \text{ for all } |\beta| = m \right\}.$$

With the norm

$$\|u\|_{C^{m, \alpha}(\overline{\Omega})} := \sum_{|\beta| \leq m} \|\partial^\beta u\|_\infty + \sum_{|\beta| = m} \text{hol}_\alpha(\partial^\beta u, \overline{\Omega}),$$

$C^{m, \alpha}(\overline{\Omega})$ is a Banach space.

Functions in $C^{0, 1}(\overline{\Omega})$ are called Lipschitz continuous with Lipschitz constant

$$\text{lip}(u, \overline{\Omega}) := \text{hol}_1(u, \overline{\Omega}).$$

A.2 L^p spaces

Definition A.5 (Banach space and dual space) Let V be a normed vector space. $(V, \|\cdot\|)$ is called complete if every Cauchy sequence converges in V . A complete normed space is called a Banach space.

The dual space of V is defined by

$$V' := \left\{ \varphi : V \rightarrow \mathbb{R} \mid \varphi \text{ linear, } \|\varphi\|_{V'} := \sup_{\|v\|=1} |\varphi(v)| < \infty \right\}.$$

The dual space of a Banach space is again a Banach space.

Definition A.6 (Operators) Let U, V be normed vector spaces and $D \subset U$. A mapping $T : D \rightarrow V$ is called an operator.

(i) T is continuous at $u \in D$ if

$$\forall \varepsilon > 0 \exists \delta > 0 \forall v \in D : \|u - v\|_U < \delta \Rightarrow \|T(u) - T(v)\|_V < \varepsilon.$$

(ii) T is continuous on D if it is continuous at every $u \in D$.

(iii) T is Lipschitz continuous if there exists $L > 0$ such that

$$\|T(u) - T(v)\|_V \leq L \|u - v\|_U \quad \forall u, v \in D.$$

Theorem A.7 (Banach fixed point theorem) Let X be a Banach space, $D \subset X$ closed, and $T : D \rightarrow D$ a contraction, i.e. Lipschitz continuous with constant $L < 1$. Then T has a unique fixed point $u \in D$.

Definition A.8 (Hilbert space) A vector space X with inner product $\langle \cdot, \cdot \rangle_X$ is called a pre-Hilbert space. A complete pre-Hilbert space is called a Hilbert space.

Definition A.9 (L^p spaces) For $p \in [1, \infty)$, $L^p(\Omega)$ denotes the space of Lebesgue measurable functions such that $|u|^p$ is integrable.

Equipped with the norm

$$\|u\|_p = \left(\int_{\Omega} |u|^p \right)^{1/p},$$

$L^p(\Omega)$ is a Banach space.

On $L^2(\Omega)$ the inner product

$$\langle u, v \rangle = \int_{\Omega} uv$$

makes $L^2(\Omega)$ a Hilbert space.

$L^\infty(\Omega)$ denotes the space of measurable functions with bounded essential supremum

$$\|u\|_\infty := \text{ess sup } |u|.$$

For $1 < p, q < \infty$ with $1/p + 1/q = 1$,

$$L^q(\Omega) \cong (L^p(\Omega))'.$$

Theorem A.10 (Hölder inequality) Let $u \in L^p(\Omega)$ and $v \in L^q(\Omega)$ with $1/p + 1/q = 1$. Then

$$\|uv\|_1 \leq \|u\|_p \|v\|_q.$$

Theorem A.11 (Young's inequality) Let $a, b > 0$ and $1/p + 1/q = 1$. Then

$$ab \leq \frac{1}{p}a^p + \frac{1}{q}b^q.$$

In particular, for $p = q = 2$ and $\varepsilon > 0$,

$$ab \leq \frac{\varepsilon}{2}a^2 + \frac{1}{2\varepsilon}b^2.$$

Theorem A.12 (Fundamental lemma of the calculus of variations) Let $u \in L^1_{\text{loc}}(\Omega) := \{u : \Omega \rightarrow \mathbb{R} \mid u \in L^1(K) \text{ for every compact set } K \subset \Omega\}$ with $\Omega \subset \mathbb{R}^n$ open. Then the following are equivalent:

$$\int_{\Omega} uv = 0 \quad \forall v \in C_0^\infty(\Omega) := \{v \in C^\infty(\Omega) \mid \text{supp}(v) \subset\subset \Omega\}$$

and

$$u = 0 \quad \text{almost everywhere in } \Omega.$$

Remark A.13 The space $C_0^\infty(\Omega)$ can be used as a class of test functions. Results obtained for such test functions extend to continuously differentiable functions that vanish on the boundary. $\triangle$

A.3 Gauss theorem and integration by parts

Definition A.14 (Lipschitz domain) An open set $\Omega \subset \mathbb{R}^d$ is called a Lipschitz domain if its boundary $\partial\Omega$ can locally be represented as the graph of a Lipschitz continuous function and Ω lies on one side of the graph.

Theorem A.15 (Gauss theorem) Let $\Omega \subset \mathbb{R}^d$ be a Lipschitz domain and

$$\mathbf{v} \in C^1(\Omega; \mathbb{R}^d) \cap C(\bar{\Omega}; \mathbb{R}^d)$$

with $\text{div}(\mathbf{v}) \in L^1(\Omega)$. Then

$$\int_{\Omega} \text{div}(\mathbf{v}) \, dx = \int_{\partial\Omega} \mathbf{v} \cdot \mathbf{n} \, d\sigma(x),$$

where $\mathbf{n}$ denotes the outward unit normal vector.

Lemma A.16 (Integration by parts) Let $\Omega \subset \mathbb{R}^d$ be a domain for which Gauss' theorem holds, $u \in C^1(\bar{\Omega})$ and $\mathbf{v} \in C^1(\bar{\Omega}; \mathbb{R}^d)$. Then

$$\int_{\Omega} \nabla u(x) \cdot \mathbf{v}(x) \, dx = - \int_{\Omega} u(x) \text{div}(\mathbf{v})(x) \, dx + \int_{\partial\Omega} u(x) \mathbf{v}(x) \cdot \mathbf{n}(x) \, d\sigma(x).$$